US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12402307
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Kim; Seungyoon et al.

Semiconductor device and data storage system including the same

Abstract

A semiconductor device and a data storage system, the device including a lower structure; and an upper structure on the lower structure and including a memory cell array, wherein the lower structure includes a semiconductor substrate, first and second active regions spaced apart from each other in a first direction on the semiconductor substrate, the first and second active regions being defined by an isolation insulating layer on the semiconductor substrate, and first and second gate pattern structures extending in the first direction to cross the first and second active regions, respectively, on the semiconductor substrate, the first gate pattern structure and the second gate pattern structure have first and second end portions spaced apart from each other in a facing manner in the first direction, respectively, and the first and second end portions are concavely curved in opposite directions away from each other in a plan view.

Inventors: **Kim; Seungyoon (Seoul, KR), Sim; Jaeryong (Suwon-si, KR), Han; Jeehoon (Hwaseong-si, KR)**

Applicant: **SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)

Family ID: **1000008780660**

Assignee: **Samsung Electronics Co., Ltd. (Suwon-si, KR)**

Appl. No.: **17/692797**

Filed: **March 11, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220406801 A1	Dec. 22, 2022

Foreign Application Priority Data

KR	10-2021-0079835	Jun. 21, 2021

Publication Classification

Int. Cl.: **H10B41/35** (20230101); **H01L23/522** (20060101); **H01L23/528** (20060101); **H10B41/10** (20230101); **H10B41/27** (20230101); **H10B43/10** (20230101); **H10B43/27** (20230101); **H10B43/35** (20230101)

U.S. Cl.:

CPC **H10B41/35** (20230201); **H01L23/5226** (20130101); **H01L23/5283** (20130101); **H10B41/10** (20230201); **H10B41/27** (20230201); **H10B43/10** (20230201); **H10B43/27** (20230201); **H10B43/35** (20230201);

Field of Classification Search

CPC: H01L (25/105); H01L (25/18); H01L (25/0657); H01L (2224/08145); H10B (41/35); H10B (41/10); H10B (41/27); H10B (41/20); H10B (41/30); H10B (41/40); H10B (41/50); H10B (43/10); H10B (43/27); H10B (43/35); H10B (43/40); H10B (43/50); H10B (43/20); H10B (43/30)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8697339	12/2013	Zhuang et al.	N/A	N/A
9012287	12/2014	Liaw	N/A	N/A
10157746	12/2017	Hsieh	N/A	N/A
10431308	12/2018	Wang	N/A	N/A
10763863	12/2019	Liaw	N/A	N/A
10790372	12/2019	Greene et al.	N/A	N/A
2008/0296695	12/2007	Yang	257/E21.409	H10D 84/85
2015/0003151	12/2014	Lee	N/A	N/A
2019/0252308	12/2018	Liu et al.	N/A	N/A
2019/0333914	12/2018	Min	N/A	H01L 21/76232
2020/0104462	12/2019	Chen et al.	N/A	N/A
2022/0367655	12/2021	Chu	N/A	H10D 30/0212
2022/0399355	12/2021	Tsuda	N/A	H10B 12/01

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20200037108	12/2019	KR	N/A
20210032894	12/2020	KR	N/A

OTHER PUBLICATIONS

Korean Intellectual Property Office Request for the Submission of an Opinion in related Korean Patent Application No. 1020210079835; mailed Apr. 9, 2025 (9 pages). cited by applicant

---

Primary Examiner: Hossain; Moazzam

Attorney, Agent or Firm: Wolf, Greenfield & Sacks, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0079835 filed on Jun. 21, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(2) Embodiments relate to a semiconductor device and a data storage system including the same.

2. Description of the Related Art

(3) A semiconductor device may be capable of storing high-capacity data in a data storage system requiring data storage. A method for increasing data storage capacity of semiconductor devices has been considered.

SUMMARY

(4) The embodiments may be realized by providing a semiconductor device including a lower structure; and an upper structure on the lower structure and including a memory cell array, wherein the lower structure includes a semiconductor substrate, a first active region and a second active region spaced apart from each other in a first direction on the semiconductor substrate, the first active region and second active region being defined by an isolation insulating layer on the semiconductor substrate, and a first gate pattern structure and a second gate pattern structure extending in the first direction to cross the first active region and the second active region, respectively, on the semiconductor substrate, the first gate pattern structure and the second gate pattern structure have a first end portion and a second end portion spaced apart from each other in a facing manner in the first direction, respectively, and the first end portion and the second end portion are concavely curved in opposite directions away from each other in a plan view.

(5) The embodiments may be realized by providing a semiconductor device including a lower structure; and an upper structure on the lower structure, wherein the lower structure includes a semiconductor substrate, a first active region and a second active region spaced apart from each other in a first direction on the semiconductor substrate, the first active region and second active region being defined by an isolation insulating layer on the semiconductor substrate, and a first gate pattern structure and a second gate pattern structure extending in the first direction to cross the first active region and the second active region, respectively, on the semiconductor substrate, the first gate pattern structure has a first side surface extending in the first direction and a first end portion facing the second gate pattern structure, the second gate pattern structure has a second side surface extending in the first direction and a second end portion facing the first end portion of the first gate pattern structure, and a minimum distance in the first direction between the first side surface of the first gate pattern structure and the second side surface of the second gate pattern structure is less than a distance in the first direction between a central portion of the first end portion of the first gate pattern structure and a central portion of the second end portion of the second gate pattern structure in a plan view.

(6) The embodiments may be realized by providing a data storage system including a semiconductor storage device including a lower structure including a semiconductor substrate, circuit elements on the semiconductor substrate, and a lower interconnection structure electrically connected to the circuit elements, an upper structure on the lower structure, and an input/output

(I/O) pad electrically connected to the circuit elements; and a controller electrically connected to the semiconductor storage device through the I/O pad and controlling the semiconductor storage device, wherein the circuit elements of the lower structure include a first active region and a second active region spaced apart from each other in a first direction on the semiconductor substrate, the first active region and second active region being defined by an isolation insulating layer on the semiconductor substrate, and a first gate pattern structure and a second gate pattern structure respectively crossing the first active region and the second active region to extend in the first direction on the semiconductor substrate, the upper structure includes an upper substrate on the lower structure, a stack structure including interlayer insulating layers and gate electrodes alternately stacked in a vertical direction, perpendicular to an upper surface of the upper substrate, on the upper substrate, and a channel structure penetrating through the stack structure in the vertical direction and including a channel layer, the first gate pattern structure and the second gate pattern structure include a first end portion and a second end portion spaced apart from each other in a facing manner in the first direction, respectively, and the first end portion and the second end portion are concavely curved in opposite directions away from each other in a plan view.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Features will be apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings in which:

(2) FIG. **1** is a schematic plan view of a semiconductor device according to example embodiments;

(3) FIG. **2**A is a schematic cross-sectional view of a semiconductor device according to example embodiments;

(4) FIG. **2**B is a schematic cross-sectional view of a semiconductor device according to example embodiments;

(5) FIG. **3** is a schematic plan view of a semiconductor device according to example embodiments;

(6) FIG. **4** is a schematic plan view of a semiconductor device according to example embodiments;

(7) FIG. **5** is a schematic plan view of a semiconductor device according to example embodiments;

(8) FIG. **6** is a schematic plan view of a semiconductor device according to example embodiments;

(9) FIG. **7** is a schematic cross-sectional view of a semiconductor device according to example embodiments;

(10) FIG. **8**A is a schematic cross-sectional view of a semiconductor device according to example embodiments;

(11) FIG. **8**B is a schematic partially enlarged cross-sectional view of a semiconductor device according to example embodiments;

(12) FIG. **9** is a schematic cross-sectional view of a semiconductor device according to example embodiments;

(13) FIGS. **10**A to **13**B are views of stages in a method of manufacturing a semiconductor device according to example embodiments;

(14) FIG. **14** is a view of a data storage system including a semiconductor device according to example embodiments;

(15) FIG. **15** is a perspective view of a data storage system including a semiconductor device according to an example embodiment; and

(16) FIG. **16** is a cross-sectional view of a semiconductor package according to an example embodiment.

DETAILED DESCRIPTION

(17) FIG. **1** is a schematic plan view of a semiconductor device according to example embodiments. In FIG. **1**, only a layout of main components of the semiconductor device is

illustrated.

(18) FIG. **2**A is a schematic cross-sectional view of a semiconductor device according to example embodiments. FIG. **2**A is a cross-sectional view of the semiconductor device of FIG. **1**, taken along line I-I′.

(19) FIG. **2**B is a schematic cross-sectional view of a semiconductor device according to example embodiments. FIG. **2**B is a cross-sectional view of the semiconductor device of FIG. **1**, taken along II-IP.

(20) Referring to FIGS. **1** to **2**B, a semiconductor device **100** may include a semiconductor substrate **201**, active regions **205**A and **205**B on the semiconductor substrate **201**, an isolation insulating layer **210** defining the active regions **205**A and **205**B, gate pattern structures GS**1** and GS**2** extending to cross the active regions **205**A and **205**B, and source/drain regions **250** in the active regions **205**A and **205**B on at least one side of the gate pattern structures GS**1** and GS**2**. The semiconductor device **100** may further include spacer layers **240**A and **240**B covering side surfaces of the gate pattern structures GS**1** and GS**2**, respectively, a buffer insulating layer **260** on the gate pattern structures GS**1** and GS**2**, an etch stop layer **270**, an insulating layer **280** on the etch stop layer **270**, and contact plugs **292** and **294**. Each of the gate pattern structures GS**1** and GS**2** may include a gate dielectric layer **212**, a lower gate pattern **214**, an upper gate pattern **220**, and a mask pattern layer **230**. The upper gate pattern **220** may include a first pattern layer **222**, a second pattern layer **224**, and a third pattern layer **226** being sequentially stacked. As used herein, the terms “first,” “second,” and the like are merely for identification and differentiation, and are not intended to imply or require sequential inclusion (e.g., a third element and a fourth element may be described without implying or requiring the presence of a first element or second element).

(21) The semiconductor substrate **201** may include a semiconductor material, e.g., a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI compound semiconductor. In an implementation, the group IV semiconductor may include silicon (Si), germanium (Ge), or silicon germanium (SiGe). The semiconductor substrate **201** may be provided as a bulk wafer, an epitaxial layer, a silicon on insulator (SOI) layer, a semiconductor on insulator (SeOI) layer, or the like. As used herein, the term “or” is not an exclusive term, e.g., “A or B” would include A, B, or A and B.

(22) The active regions **205**A and **205**B may be defined by the isolation insulating layer **210** in the semiconductor substrate **201** and may be spaced apart from each other in a first direction, e.g., X-direction. The active regions **205**A and **205**B may include portions crossing the gate pattern structures GS**1** and GS**2** and extending in a second direction, e.g., Y-direction. Upper end portions of the active regions **205**A and **205**B may be positioned on a level lower than an upper surface of the isolation insulating layer **210**. The source/drain regions **250** including impurities may be in portions of the active regions **205**A and **205**B. Accordingly, the active regions **205**A and **205**B may be understood as regions including regions in which the source/drain regions **250** are disposed.

(23) At least one of the active regions **205**A and **205**B may include an extension portion extending (e.g., lengthwise) in the Y-direction, and contact portions on both sides of the extension portion in the Y-direction and connected to the extension portion, and the contact portion may be bent from the extension portion. In an implementation, the first active region **205**A may include a first extension portion and a first contact portion and a second contact portion disposed on both sides of the first extension portion and bent from the first extension portion in opposite directions. The first contact portion and the second contact portion may be part of regions in which the source/drain regions **250** are disposed.

(24) The isolation insulating layer **210** may define the active regions **205**A and **205**B in the semiconductor substrate **201**. The isolation insulating layer **210** may be formed by, e.g., a shallow trench isolation (STI) process. The isolation insulating layer **210** may cover side surfaces of the active regions **205**A and **205**B facing each other in the X-direction. An upper surface of the isolation insulating layer **210** between the first active region **205**A and the second active region

205B may have a region RA, which is concave in a direction toward a lower surface of the isolation insulating layer **210**. The isolation insulating layer **210** may be formed of an insulating material. The isolation insulating layer **210** may include, e.g., silicon oxide, silicon nitride, or a combination thereof.

(25) In an implementation, a distance da (e.g., maximum distance) between the first active region **205**A and the second active region **205**B in the first direction, e.g., X-direction, may be about 160 nm or less. The example embodiment may effectively improve or address issues or problems that could arise when a distance between end portions PA**1** and PA**2** of the gate pattern structures decreases as the patterns of the active regions **205**A and **205**B are miniaturized below the range mentioned above. The distance da may be greater than 0 nm and may be, e.g., greater than minimal pattern spacing achievable with photolithography equipment. It may also be understood that a width (e.g., maximum width) of the isolation insulating layer **210** between the first active region **205**A and the second active region **205**B in the X-direction is about 160 nm or less.

(26) The gate pattern structures GS**1** and GS**2** may include a first gate pattern structure GS**1** and a second gate pattern structure GS**2** spaced apart from each other in the first direction, e.g., X-direction. The first gate pattern structure GS**1** may cross the first active region **205**A and extend in the X-direction, and the second gate pattern structure GS**2** may cross the second active region **205**B and extend in the X-direction. The first gate pattern structure GS**1** may have the first end portion PA**1** facing a gate isolation region CT, and the second gate pattern structure GS**2** may have the second end portion PA**2** facing the gate isolation region CT. The first end portion PA**1** and the second end portion PA**2** may face each other in the X-direction and may be spaced apart from each other. In an implementation, the first end portion PA**1** may face the second gate pattern structure GS**2**, and the second end portion PA**2** may face the first end portion PA**1** of the first gate pattern structure GS**1**. The gate isolation region CT may indicate a region in which gate pattern structures on the same straight line in the X-direction are separated from each other.

(27) In an implementation, in a plan view, the first end portion PA**1** and the second end portion PA**2** facing each other may be concavely curved in opposite directions away from each other (e.g., may have concavities that are open toward one another).

(28) In an implementation, the first gate pattern structure GS**1** may have a first side surface extending in the X-direction, and the second gate pattern structure GS**2** may have a second side surface extending in the X-direction. In a plan view, a first distance D**1** (in the X direction) between the first side surface of the first gate pattern structure GS**1** and the second side surface of the second gate pattern structure GS**2** may be less than a second distance D**2** between a central portion of the first end portion PA**1** of the first gate pattern structure GS**1** and a central portion of the second end portion PA**2** of the second gate pattern structure GS**2**. The first distance D**1** may be a minimum distance between the first side surface of the first gate pattern structure GS**1** and the second side surface of the second gate pattern structure GS**2**.

(29) In an implementation, in a plan view, at an edge or a corner of the first end portion PA**1**, the first side surface and a cross-section of the first end portion PA**1** of the first gate pattern structure GS**1** may meet at an acute angle ($0°<\alpha<90°$). In a plan view, the side surfaces may also meet at an acute angle at another corner of the first end portion PA**1** and both side corners of the second end portion PA**2**.

(30) As the patterns of semiconductor devices shrink, the distance between the gate pattern structures may be reduced. Accordingly, the end portions of the gate pattern structures could be convexly rounded to reduce the area of a region in which the gate pattern structure and the active region intersect, thereby causing device failure. In addition, as the distance between the end portions of the adjacent gate pattern structures decreases, a patterning defect in which the gate pattern structures are not separated from each other could also occur, thereby causing device failure due to a short circuit between the gate pattern structures. According to an example embodiment, the gate isolation region CT may be in a rounded hole region to separate the gate pattern structures

GS**1** and GS**2**, and the end portions PA**1** of the gate pattern structures GS**1** and GS**2** may be concavely rounded. Accordingly, device defects due to convex rounding of the end portions may be minimized, and a distance between the end portions may be secured to help reduce or prevent short circuits between the gate patterns.

(31) A gate dielectric layer **212** may be on the active regions **205**A and **205**B. In an implementation, a side surface of the gate dielectric layer **212** may be substantially coplanar with a side surface of the first active region **205**A. The gate dielectric layer **212** may be formed of silicon oxide.

(32) The lower gate pattern **214** may be on the gate dielectric layer **212**. A side surface of the lower gate pattern **214** may be substantially coplanar with a side surface of the gate dielectric layer **212**. The lower gate pattern **214** may include a semiconductor layer including, e.g., polycrystalline silicon.

(33) As used herein, the description of an element being substantially coplanar with another element refers to a case of being coplanar or of having a difference in the range of deviations occurring during a manufacturing process and may be interpreted as having the same meaning even without the expression “substantially.”

(34) The upper gate pattern **220** may be on the lower gate pattern **214**. The upper gate pattern **220** may extend longer in the X-direction than the side surface of the lower gate pattern **214**. At least a portion of the upper gate pattern **220**, e.g., the first pattern layer **222** of the upper gate pattern **220**, may cover a portion of the side surface of the lower gate pattern **214** and may extend onto an upper surface of the isolation insulating layer **210**. The first pattern layer **222** of the upper gate pattern **220** may include a semiconductor layer including, e.g., polycrystalline silicon. The second pattern layer **224** of the upper gate pattern **220** may be a barrier layer, and the barrier layer may include a metal nitride or a metal silicon nitride. The metal nitride may include, e.g., titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN), and the metal silicon nitride may include, e.g., titanium silicon nitride (TiSiN), tantalum silicon nitride (TaSiN), or tungsten silicon nitride (WSiN). In an implementation, the second pattern layer **224** may include graphene. The third pattern layer **226** of the upper gate pattern **220** may include a metal layer including, e.g., tungsten (W), copper (Cu), aluminum (Al), molybdenum (Mo), or rubidium (Rb).

(35) The mask pattern layer **230** may be on the upper gate pattern **220**. The mask pattern layer **230** may include, e.g., silicon nitride, silicon oxynitride, silicon carbonitride, or silicon oxide.

(36) Herein, the components constituting the first gate pattern structure GS**1** may be referred to as “the first gate dielectric layer **212**, the first lower gate pattern **214**, the first upper gate pattern **220**, and the first mask pattern layer **230**,” respectively, and the components constituting the second gate pattern structure GS**2** may be referred to as “the second gate dielectric layer **212**, the second lower gate pattern **214**, the second upper gate pattern **220**, and the second mask pattern layer **230**,” respectively.

(37) The spacer layers **240**A and **240**B may be on side surfaces of the gate pattern structures GS**1** and GS**2**. The spacer layers **240**A and **240**B may include a first spacer layer **240**A, surrounding side surfaces of the first gate pattern structure GS**1** and a second spacer layer **240**B surrounding side surfaces of the second gate pattern structure GS**2**. The first spacer layer **240**A may cover the first side surface and the first end portion PA**1** of the first gate pattern structure GS**1**. The second spacer layer **240**B may cover the second side surface and the second end portion PA**2** of the second gate pattern structure GS**2** and may be spaced apart from the first spacer layer **240**A. In the gate isolation region CT, the first spacer layer **240**A and the second spacer layer **240**B may cover the concave region RA of the upper surface of the isolation insulating layer **210**. The spacer layers **240**A and **240**B may be formed along side profiles of the end portions PA**1** and PA**2** of the gate pattern structures GS**1** and GS**2**. In an implementation, the spacer layers **240**A and **240**B may have a shape with one surface concavely curved toward the gate pattern structures GS**1** and GS**2** in a plan view. The spacer layers **240**A and **240**B may insulate the gate pattern structures GS**1** and GS**2**

from the source/drain regions **250**. The spacer layers **240**A and **240**B may be formed of silicon oxide, silicon nitride, or silicon oxynitride, and may include a plurality of layers.
(38) The source/drain regions **250** may be in the active regions **205**A and **205**B on or at both sides of the gate pattern structures GS**1** and GS**2**, respectively. The source/drain regions **250** may serve as a source region or a drain region of the transistor. The source/drain regions **250** may include P-type or N-type impurities. The source/drain regions **250** may include a plurality of regions including elements of different concentrations or doped elements.
(39) The buffer insulating layer **260** may be on the gate pattern structures GS**1** and GS**2**. The buffer insulating layer **260** may cover the first spacer layer **240**A, the second spacer layer **240**B, the source/drain regions **250**, and the mask pattern layer **230**. A portion of the buffer insulating layer **260** may extend downwardly along outer surfaces of the spacer layers **240**A and **240**B in the gate isolation region CT to cover the concave region RA of the upper surface of the isolation insulating layer **210**. The buffer insulating layer **20** may be formed of an insulating material, e.g., silicon oxide, silicon nitride, or silicon oxynitride.
(40) The etch stop layer **270** may be on the buffer insulating layer **260**. The etch stop layer **270** may extend between the first gate pattern structure GS**1** and the second gate pattern structure GS**2** and may be sharp toward an upper surface of the isolation insulating layer **210** between the first active region **205**A and the second active region **205**B. The etch stop layer **270** may be formed of an insulating material, and may be formed of a material different from that of the insulating layer **280**. The etch stop layer **270** may be formed of, e.g., silicon oxide, silicon nitride, or silicon oxynitride.
(41) The insulating layer **280** may be on the etch stop layer **270**. The insulating layer **280** may be formed of an insulating material. The insulating layer **280** may include a plurality of insulating layers.
(42) The contact plugs **292** and **294** may penetrate through the insulating layer **280**. First contact plugs **292**, among the contact plugs **292** and **294**, may be connected to the source/drain regions **250** through the etch stop layer **270** and the buffer insulating layer **260**. The second contact plugs **294**, among the contact plugs **292** and **294**, may be connected to the upper gate pattern **220** through the etch stop layer **270**, the buffer insulating layer **260**, and the mask pattern layer **230**. The contact plugs **292** and **294** may include a conductive material, e.g., tungsten (W), copper (Cu), aluminum (Al), or the like, and may further include a barrier layer formed of a metal nitride.
(43) FIGS. **3** to **5** are schematic plan views of semiconductor devices according to example embodiments.
(44) Referring to FIG. **3**, shapes of active regions **205**A_**1** and **205**B_**1** of the semiconductor device **100**A may vary. At least one of the active regions **205**A_**1** and **205**B_**1** may include an extension portion extending in the Y-direction and first and second contact portions on both sides of the extension portion in the Y-direction. The first contact portion and the second contact portion may be bent from the extension portion in the same direction, e.g., in the X-direction.
(45) Referring to FIG. **4**, shapes of active regions **205**A_**2** and **205**B_**2** of the semiconductor device **100**B may vary. At least one of the active regions **205**A_**2** and **205**B_**2** may have an ‘I’ shape or a shape similar thereto.
(46) Referring to FIG. **5**, at least one of active regions **205**A_**3** and **205**B_**3** of a semiconductor device **100**C may extend in a direction oblique to the X-direction. In an implementation, the first active region **205**A_**3** may extend in a direction oblique (e.g., inclined) to the X and Y-directions and may have a parallelogram shape or a shape similar thereto in a plan view.
(47) FIG. **6** is a schematic plan view of a semiconductor device according to example embodiments.
(48) Referring to FIG. **6**, in a semiconductor device **100**D, the gate isolation regions CTa and CTb separating the gate pattern structures in the X-direction may be formed in an irregular arrangement or shape. In an implementation, first and second active regions **205**Aa and **205**Ba may be spaced apart from each other by a first interval S**1** in the X-direction, and third and fourth active regions

205 Ab and **205**Bb may be spaced apart from each other by a second interval S**2** (greater than the first interval S**1**). Even if the intervals between the active regions are different, the gate isolation regions CTa and CTb may be formed in a non-continuous rounded hole type, so that the gate pattern structures may be less affected by the arrangement of the patterns in the same straight line than a case in which the gate isolation region is formed in a line shape, and thus, the gate pattern structures may be more stably separated.
(49) FIG. **7** is a schematic cross-sectional view of a semiconductor device according to example embodiments.
(50) Referring to FIG. **7**, a semiconductor device **300** may include a lower structure **100′** and an upper structure **200** on the lower structure **100′**. The lower structure **100′** may correspond to the structure including the semiconductor substrate **201**, the active regions **205**, the gate pattern structure GS, and the source/drain regions **250** of the semiconductor device of FIGS. **1** to **6** described above. The lower structure **100′** may include a lower interconnection structure **290** including contact plugs **292** and **294** and lower interconnections **296**. A memory cell region CELL may be in the upper structure **200**, and the memory cell region CELL may include nonvolatile memory devices such as DRAM, static RAM (SRAM), or the like, and volatile memory devices such as PRAM, MRAM, ReRAM, or flash memory device. In an implementation, the lower structure **100′** and the upper structure **200** may be arranged side by side in a horizontal direction.
(51) FIG. **8**A is a schematic cross-sectional view of a semiconductor device according to example embodiments.
(52) FIG. **8**B is a schematic partially enlarged cross-sectional view of a semiconductor device according to example embodiments. FIG. **8**B is an enlarged view of region ‘A’ of FIG. **8**A.
(53) Referring to FIGS. **8**A and **8**B, a semiconductor device **300**A may include a lower structure **100′** and an upper structure **200**, and a memory cell region of the upper structure **200** may include a flash memory device including gate electrodes **130** and a channel structure CH. The lower structure **100′** may correspond to a structure including circuit elements including the semiconductor substrate **201**, the active regions **205**, the gate pattern structure GS, and the source/drain regions **250** of the semiconductor device of FIGS. **1** to **6** described above.
(54) The upper structure **200** may include an upper substrate **101**, interlayer insulating layers **120** and gate electrodes **130** alternately stacked on the upper substrate **101** in the Z-direction, a channel structure CH penetrating through the gate electrodes **130** in the Z-direction and including a channel layer **140**, and an upper interconnection structure **170**. The upper structure **200** may further include first and second horizontal conductive layers **104** and **105** between the upper substrate **101** and a stack structure of the gate electrodes **130**, a horizontal insulating layer **110**, an upper insulating layer **190**, gate contact plugs **162**, a source contact plug **164**, and a peripheral contact plug **167**.
(55) The upper substrate **101** may include a semiconductor material, e.g., a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI compound semiconductor. The upper substrate **101** may include, e.g., a polycrystalline silicon layer having N-type or P-type conductivity. The upper substrate **101** may be electrically connected to the upper interconnection structure **170** through the source contact plug **164**.
(56) The gate electrodes **130** may be stacked on the upper substrate **101** and spaced apart from each other in the Z-direction. The gate electrodes **130** may extend in the Y-direction. The gate electrodes **130** may include a lower gate electrode forming a gate of a ground select transistor, memory gate electrodes forming a plurality of memory cells, and upper gate electrodes forming the gates of string select transistors. The number of memory gate electrodes constituting the memory cells may be determined according to capacity of the semiconductor device **300**A. In an implementation, each of the gate electrodes **130** constituting the string select transistor and the ground select transistor may be one or two or more.
(57) The gate electrodes **130** may be vertically spaced apart and stacked on the upper substrate **101**, and may extend to have different lengths in the X-direction to form a step structure in the form of a

step. Due to the step structure, the gate electrodes **130** may have pad regions in which the lower gate electrode **130** extends longer than the upper gate electrode **130** so as to be exposed upwardly, and gate contact plugs **162** may be disposed in the pad regions and connected to the gate electrodes **130**. The gate contact plugs **162** may be electrically connected to the lower interconnection structure **290** of the lower structure **100′** through through-contact plugs penetrating through a separate through-region.

(58) The gate electrodes **130** may each include a first layer and a second layer. The first layer may cover upper and lower surfaces of the second layer and may extend between the channel structures CH and the second layer. The first layer may include a high dielectric material such as aluminum oxide (AlO), and the second layer may include titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), tungsten (W), or tungsten nitride (WN). In an implementation, the gate electrodes **130** may include polycrystalline silicon or a metal-semiconductor compound.

(59) The interlayer insulating layers **120** may be between the gate electrodes **130**. Like the gate electrodes **130**, the interlayer insulating layers **120** may be spaced apart from each other in the Z-direction and may be disposed to extend in the Y-direction. The interlayer insulating layers **120** may include an insulating material such as silicon oxide. Some of the interlayer insulating layers **120** may have different thicknesses.

(60) The first horizontal conductive layer **104** may function a portion of a common source line of the semiconductor device **300**A, e.g., as a common source line together with the upper substrate **101**. The first horizontal conductive layer **104** may not extend to a region in which the gate electrodes **130** form a step structure. The first horizontal conductive layer **104** and the second horizontal conductive layer **105** may include a conductive material, e.g., doped polycrystalline silicon. In this case, at least the first horizontal conductive layer **104** may be a layer doped with impurities of the same conductivity type as that of the upper substrate **101**, and the second horizontal conductive layer **105** may be a doped layer or a layer including impurities spread from the first horizontal conductive layer **104**. In an implementation, a material of the second horizontal conductive layer **105** may be replaced with an insulating layer.

(61) The horizontal insulating layer **110** may include first to third horizontal insulating layers **111**, **112**, and **113** stacked on the upper substrate **101**. The horizontal insulating layers **111**, **112**, and **113** may be partially replaced with the first horizontal conductive layer **104** of FIG. **8**A through a subsequent process. The first and third horizontal insulating layers **111** and **113** may be formed of the same material, and the second horizontal insulating layer **112** may be formed of a material different from that of the first and third horizontal insulating layers **111** and **113**. The second horizontal conductive layer **105** may be bent, while covering side surfaces of the horizontal insulating layers **111**, **112**, and **113** in a region in which the horizontal insulating layers **111**, **112**, and **113** are patterned, so as to be in contact with the upper substrate **101**.

(62) The channel structures CH may each form one memory cell string, and may be spaced apart from each other, while forming rows and columns. The channel structures CH may form a grid pattern or may be disposed in a zigzag form in one direction. The channel structures CH may penetrate through the stack structure of the gate electrodes **130** in the Z-direction. The channel structures CH may have a columnar shape and may have inclined side surfaces narrower in width in a direction toward the upper substrate **101** according to an aspect ratio.

(63) The channel layer **140** may be in the channel structures CH. In the channel structures CH, the channel layer **140** may have an annular shape surrounding an internal core insulating layer **150**. The channel layer **140** may be in contact with the first horizontal conductive layer **104** from a lower portion thereof and may be electrically connected to the upper substrate **101**. The channel layer **140** may include a semiconductor material such as polycrystalline silicon or single crystal silicon.

(64) A channel pad **155** may be on the channel layer **140** in the channel structures CH. The channel pad **155** may cover an upper surface of the core insulating layer **147** and be electrically connected

to the channel layer **140**. The channel pad **155** may include, e.g., doped polycrystalline silicon. The channel pad **155** may include a semiconductor material such as polycrystalline silicon or single crystal silicon and may include, e.g., doped polycrystalline silicon.
(65) A gate dielectric layer **145** may be between the gate electrodes **130** and the channel layer **140**. The gate dielectric layer **145** may include a tunneling layer, an information storage layer, and a blocking layer sequentially stacked from the channel layer **140**. The tunneling layer may tunnel charges into the information storage layer and may include, e.g., silicon oxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon oxynitride (SiON), or a combination thereof. The information storage layer may be a charge trap layer or a floating gate conductive layer. The blocking layer may include silicon oxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon oxynitride (SiON), a high-k dielectric material, or a combination thereof. In an implementation, at least a portion of the gate dielectric layer **145** may extend along the gate electrodes **130** in a horizontal direction.
(66) As shown in FIG. **8**B, the channel structures CH may penetrate through the gate electrodes **130** in the Z-direction and penetrate through the first horizontal conductive layer **104** and the second horizontal conductive layer **105** in the Z-direction so as to extend into the upper substrate **101**. The first horizontal conductive layer **104** may be in contact with the channel layer **140** through the gate dielectric layer **145**. The first horizontal conductive layer **104** may be directly connected to the channel layer **140** around the channel layer **140**.
(67) The gate contact plugs **162** may be connected to the gate electrodes **130**, the source contact plug **164** may be connected to the upper substrate **101**, and the peripheral contact plug **167** may be connected to the lower interconnection structure **290**. The gate contact plugs **162**, the source contact plug **164**, and the peripheral contact plug **167** may include a conductive pattern and a barrier layer covering side surfaces and bottom surface of the conductive pattern. The barrier layer may include, e.g., titanium (Ti), titanium nitride (TiN), tantalum (Ta), or tantalum nitride (TaN). The conductive pattern may include a metal material, e.g., tungsten (W), titanium (Ti), copper (Cu), cobalt (Co), aluminum (Al), or alloys thereof.
(68) The upper interconnection structure **170** may include first contacts **172**, second contacts **174**, and upper interconnections **176**. The upper interconnection structure **170** may be electrically connected to the gate electrodes **130** and the channel structures CH. The first contacts **172** may be on the channel pads **155** and the gate contact plugs **162**, and the second contacts **174** may be on the first contacts **172**. The upper interconnection structure **170** may include a conductive pattern and a barrier layer covering side surfaces and a bottom surface of the conductive pattern. The barrier layer may include, e.g., titanium (Ti), titanium nitride (TiN), tantalum (Ta), or tantalum nitride (TaN). The conductive pattern may include a metal material, e.g., tungsten (W), titanium (Ti), copper (Cu), cobalt (Co), aluminum (Al), or alloys thereof. The number of layers and arrangement of contacts and upper interconnections constituting the upper interconnection structure **170** may be variously changed.
(69) FIG. **9** is a schematic cross-sectional view of a semiconductor device according to example embodiments.
(70) Referring to FIG. **9**, a lower structure **100**′ and an upper structure **200** of a semiconductor device **300**B may be bonded to each other through a bonding structure. The upper structure **200** of the semiconductor device **300**B is shown by inverting the upper structure **200** of the semiconductor device **300**A of FIG. **8**A, the lower structure **100**′ may further include a lower bonding structure **310**, and the upper structure **200** may further include an upper bonding structure **320**. The peripheral contact plug **167** may be connected to a pad layer **340** formed in an opening of an outer insulating layer **330** disposed on the upper substrate **101**.
(71) The lower bonding structure **310** may include a lower bonding via **312** and a lower bonding pad **314** electrically connected to the lower interconnection structure **290**. The upper bonding structure **320** may include an upper bonding via **322** and an upper bonding pad **324** electrically connected to the upper interconnection structure **170**. The lower bonding structure **310** and the

upper bonding structure **320** may each include, e.g., tungsten (W), aluminum (Al), copper (Cu), tungsten nitride (WN), tantalum nitride (TaN), and titanium nitride (TiN), or a combination thereof. The lower bonding pad **314** and the upper bonding pad **324** may function as bonding layers for bonding the lower structure **100**′ and the upper structure **200**. In an implementation, the lower bonding pad **314** and the upper bonding pad **324** may provide an electrical connection path between the lower structure **100**′ and the upper structure **200**. The lower bonding pad **314** and the upper bonding pad **324** may be bonded by copper (Cu)-to-copper (Cu) bonding.
(72) FIGS. **10**A to **13**B are views of stages in a method of manufacturing a semiconductor device according to example embodiments. FIGS. **10**A, **11**A, **12**A, and **13**A each are plan views according to a process sequence of a region corresponding to FIG. **1**, and FIGS. **10**B, **11**B, **12**B, and **13**B each are cross-sectional views according to a process sequence of a region corresponding to FIGS. **2**A and **2**B.
(73) Referring to FIGS. **10**A and **10**B, the semiconductor substrate **201**, the gate dielectric layer **212**, and the lower gate pattern **214** may be patterned to define active regions **205**A and **205**B. Material layers forming the gate dielectric layer **212** and the lower gate pattern **214** may be deposited on the semiconductor substrate **201** and an etching process may be performed thereon using a mask layer to form trenches, thereby forming the active regions **205**A and **205**B. The trenches may have a shape with a width narrowing downwardly. In an implementation, the active regions **205**A and **205**B may be patterned to have a shape as shown in FIG. **10**A, or may be patterned to have various suitable shapes.
(74) Referring to FIGS. **11**A and **11**B, an upper gate pattern **220** and a mask pattern layer **230** may be formed on the active regions **205**A and **205**B.
(75) The upper gate pattern **220** and the mask pattern layer **230** may be formed to have a line shape that crosses the active regions **205**A and **205**B and extends in the X-direction. In an implementation, a spin on hardmask (SOH) mask layer and an antireflection layer including SiON may be formed on the mask pattern layer **230**, and an etching process may then be performed. On both sides of the upper gate pattern **220** and the mask pattern layer **230**, the gate dielectric layer **212** and the lower gate pattern **214** formed on the active regions **205**A and **205**B may be partially removed.
(76) Referring to FIGS. **12**A and **12**B, a gate separation hole E_CT may be formed.
(77) Between the active regions **205**A and **205**B, the gate dielectric layer **212**, the lower gate pattern **214**, the upper gate pattern **220**, and the mask pattern layer **230** may be etched to form the gate separation hole E_CT. The first gate pattern structure GS**1** and the second gate pattern structure GS**2** (separated from each other in the X-direction) may be formed or defined by the gate separation hole E_CT. The gate separation hole E_CT may have a rounded hole shape instead of a line shape, and accordingly, the gate pattern structures GS**1** and GS**2** may be formed to have concavely (e.g., inwardly) rounded end portions PA**1** and PA**2**. The gate separation hole E_CT may partially recess an upper surface of the isolation insulating layer **210**.
(78) Referring to FIGS. **13**A and **13**B, spacer layers **240**A and **240**B may be formed, and source/drain regions **250** may be formed.
(79) The spacer layers **240**A and **240**B may be formed on side surfaces of the gate pattern structures GS**1** and GS**2**. The spacer layers **240**A and **240**B may be formed to cover the end portions PA**1** and PA**2** of the gate pattern structures GS**1** and GS**2**.
(80) On both sides of the gate pattern structures GS**1** and GS**2**, source/drain regions **250** may be formed by implanting impurities into the active regions **205**A and **205**B.
(81) Next, referring to FIGS. **1** to **2**B, after the buffer insulating layer **260**, the etch stop layer **270**, and the insulating layer **280** are formed, the contact plugs **292** and **294** may be formed. Accordingly, the semiconductor device **100** of FIGS. **1** to **2**B may be manufactured.
(82) Meanwhile, after the lower interconnections **296** connected to the contact plugs **292** and **294** are formed to form the lower structure **100**′ including the lower interconnection structure **290**, the

upper structure **200** may be further formed to manufacture the semiconductor device **300**. In an implementation, by alternately stacking gate electrodes **130** and interlayer insulating layers **120** on the upper substrate **101** and forming channel structures CH penetrating therethrough, the semiconductor device **300**A shown in FIGS. **8**A and **8**B. may also be manufactured.

(83) FIG. **14** is a view schematically illustrating a data storage system including a semiconductor device according to example embodiments.

(84) Referring to FIG. **14**, a data storage system **1000** may include a semiconductor device **1100** and a controller **1200** electrically connected to the semiconductor device **1100**. The data storage system **1000** may be a storage device including one or a plurality of semiconductor devices **1100** or an electronic device including a storage device. In an implementation, the data storage system **1000** may be a solid state drive (SSD) device including one or a plurality of semiconductor devices **1100**, a universal serial bus (USB), a computing system, a medical device, or a communication device.

(85) The semiconductor device **1100** may be a non-volatile memory device and may be, e.g., the NAND flash memory device described above with reference to FIGS. **1** to **9**. The semiconductor device **1100** may include a first structure **1100**F and a second structure **1100**S on the first structure **1100**F. In an implementation, the first structure **1100**F may be next to the second structure **1100**S. The first structure **1100**F may be a peripheral circuit structure including a decoder circuit **1110**, a page buffer **1120**, and a logic circuit **1130**. The second structure **1100**S may be a memory cell structure including a bit line BL, a common source line CSL, word lines WL, first and second gate upper lines UL**1** and UL**2**, and first and second gate lower lines LL**1** and LL**2**, and memory cell strings CSTR between the bit line and the common source line CSL.

(86) In the second structure **1100**S, each of the memory cell strings CSTR may include lower transistors LT**1** and LT**2** adjacent to the common source line CSL, upper transistors UT**1** and UT**2** adjacent to the bit line BL, a plurality of memory cell transistors MCT disposed between the lower transistors LT**1** and LT**2** and the upper transistors UT**1** and UT**2**. The number of lower transistors LT**1** and LT**2** and the number of upper transistors UT**1** and UT**2** may be variously modified according to embodiments.

(87) In an implementation, the upper transistors UT**1** and UT**2** may include a string select transistor, and the lower transistors LT**1** and LT**2** may include a ground select transistor. The gate lower lines LL**1** and LL**2** may be gate electrodes of the lower transistors LT**1** and LT**2**, respectively. The word lines WL may be gate electrodes of the memory cell transistors MCT, and the gate upper lines UL**1** and UL**2** may be gate electrodes of the upper transistors UT**1** and UT**2**, respectively.

(88) In an implementation, the lower transistors LT**1** and LT**2** may include a lower erase control transistor LT**1** and a ground select transistor LT**2** connected in series. The upper transistors UT**1** and UT**2** may include a string select transistor UT**1** and an upper erase control transistor UT**2** connected in series. At least one of the lower erase control transistor LT**1** and the upper erase control transistor UT**1** may be used for an erase operation of erasing data stored in the memory cell transistors MCT using a gate induced drain leakage (GIDL) phenomenon.

(89) The common source line CSL, the first and second gate lower lines LL**1** and LL**2**, the word lines WL, and the first and second gate upper lines UL**1** and UL**2** may be electrically connected to a decoder circuit **1110** through first connection lines **1115** extending from within the first structure **1100**F to the second structure **1100**S. The bit lines BL may be electrically connected to the page buffer **1120** through second connection wires **1125** extending from within the first structure **1100**F to the second structure **1100**S.

(90) In the first structure **1100**F, the decoder circuit **1110** and the page buffer **1120** may perform a control operation on at least one selected memory cell transistor among the plurality of memory cell transistors MCT. The decoder circuit **1110** and the page buffer **1120** may be controlled by the logic circuit **1130**. The semiconductor device **1100** may communicate with the controller **1200** through an input/output (I/O) pad **1101** electrically connected to the logic circuit **1130**. The I/O pad **1101** may be electrically connected to the logic circuit **1130** through an I/O connection line **1135**

extending from within the first structure **1100**F to the second structure **1100**S.
(91) The controller **1200** may include a processor **1210**, a NAND controller **1220**, and a host interface **1230**. In an implementation, the data storage system **1000** may include a plurality of semiconductor devices **1100**, and in this case, the controller **1200** may control the plurality of semiconductor devices **1100**.
(92) The processor **1210** may control an overall operation of the data storage system **1000** including the controller **1200**. The processor **1210** may operate according to a predetermined firmware and may access the semiconductor device **1100** by controlling the NAND controller **1220**. The NAND controller **1220** may include a NAND interface **1221** that handles communication with the semiconductor device **1100**. Through the NAND interface **1221**, a control command for controlling the semiconductor device **1100**, data to be written into the memory cell transistors MCT of the semiconductor device **1100**, and data to be read from the memory cell transistors may be transmitted. The host interface **1230** may provide a communication function between the data storage system **1000** and an external host. When a control command is received from an external host through the host interface **1230**, the processor **1210** may control the semiconductor device **1100** in response to the control command.
(93) FIG. **15** is a perspective view schematically illustrating a data storage system including a semiconductor device according to an example embodiment.
(94) Referring to FIG. **15**, a data storage system **2000** according to an example embodiment may include a main substrate **2001**, a controller **2002** mounted on the main substrate **2001**, one or more semiconductor packages **2003**, and a DRAM **2004**. The semiconductor package **2003** and the DRAM **2004** may be connected to the controller **2002** by interconnection patterns **2005** on the main substrate **2001**.
(95) The main substrate **2001** may include a connector **2006** including a plurality of pins coupled to an external host. The number and arrangement of the plurality of pins in the connector **2006** may vary according to a communication interface between the data storage system **2000** and the external host. In an implementation, the data storage system **2000** may communicate with an external hot according to any one of interfaces such as M-Phy for a universal serial bus (USB), a peripheral component interconnect express (PCI-express), a serial advanced technology attachment (SATA), a universal flash storage (UFS), or the like. In an implementation, the data storage system **2000** may be operated by power supplied from the external host through the connector **2006**. The data storage system **2000** may further include a power management integrated circuit (PMIC) for distributing power supplied from the external host to the controller **2002** and the semiconductor package **2003**.
(96) The controller **2002** may write data to the semiconductor package **2003** or read data therefrom and may improve an operating rate of the data storage system **2000**.
(97) The DRAM **2004** may be a buffer memory for alleviating a speed difference between the semiconductor package **2003**, which is a data storage space, and the external host. The DRAM **2004** included in the data storage system **2000** may operate as a kind of cache memory and may provide a space for temporarily storing data in a control operation for the semiconductor package **2003**. When the data storage system **2000** includes the DRAM **2004**, the controller **2002** may further include a DRAM controller for controlling the DRAM **2004** in addition to the NAND controller for controlling the semiconductor package **2003**.
(98) The semiconductor package **2003** may include first and second semiconductor packages **2003***a* and **2003***b* spaced apart from each other. Each of the first and second semiconductor packages **2003***a* and **2003***b* may be a semiconductor package including a plurality of semiconductor chips **2200**. Each of the first and second semiconductor packages **2003***a* and **2003***b* may include a package substrate **2100**, the semiconductor chips **2200** on the package substrate **2100**, adhesive layers **2300** on lower surfaces of the semiconductor chips **2200**, respectively, a connection structure **2400** electrically connecting the semiconductor chips **2200** and the package substrate **2100** to each other, and a molding layer **2500** covering the semiconductor chips **2200** and the connection

structure **2400** on the package substrate **2100**.
(99) The package substrate **2100** may be a printed circuit board including package upper pads **2130**. Each semiconductor chip **2200** may include an I/O pad **2210**. The I/O pad **2210** may correspond to the I/O pad **1101** of FIG. **14**. Each of the semiconductor chips **2200** may include gate stack structures **3210** and channel structures **3220**. Each of the semiconductor chips **2200** may include the semiconductor device described above with reference to FIGS. **1** to **9**.
(100) In an implementation, the connection structure **2400** may be a bonding wire electrically connecting the I/O pad **2210** and the package upper pads **2130** to each other. Accordingly, in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a bonding wire method and may be electrically connected to the package upper pads **2130** of the package substrate **2100**. In an implementation, in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a connection structure including a through-electrode (i.e., through-silicon via (TSV)), instead of the bonding wire-type connection structure **2400**.
(101) In an implementation, the controller **2002** and the semiconductor chips **2200** may be included in a single package. In an implementation, the controller **2002** and the semiconductor chips **2200** may be mounted on a separate interposer substrate different from the main substrate **2001** and may be connected to each other by an interconnection on the interposer substrate.
(102) FIG. **16** is a cross-sectional view schematically illustrating a semiconductor package according to an example embodiment. FIG. **16** illustrates an example embodiment of the semiconductor package **2003** of FIG. **15** and conceptually illustrates a region of the semiconductor package **2003** of FIG. **15**, taken along line IV-IV′.
(103) Referring to FIG. **16**, in the semiconductor package **2003**, the package substrate **2100** may be a printed circuit board. The package substrate **2100** may include a package substrate body portion **2120**, the package upper pads **2130** (refer to FIG. **15**) on an upper surface of the package substrate body portion **2120**, lower pads **2125** on a lower surface of the package substrate body portion **2120** or exposed through the lower surface thereof, and internal interconnections **2135** electrically connecting the package upper pads **2130** and the lower pads **2125** to each other in the package substrate body portion **2120**. The package upper pads **2130** may be electrically connected to the connection structures **2400**. The lower pads **2125** may be connected to the interconnection patterns **2005** of the main substrate **2010** of the data storage system **2000** as shown in FIG. **15** through conductive connectors **2800**.
(104) Each of the semiconductor chips **2200** may include a semiconductor substrate **3010** and a first structure **3100** and a second structure **3200** sequentially stacked on the semiconductor substrate **3010**. The first structure **3100** may include a peripheral circuit region including peripheral interconnections **3110**. The second structure **3200** may include a common source line **3205**, a gate stack structure **3210** on the common source line **3205**, channel structures **3220** and isolation regions **3230** penetrating through the gate stack structure **3210**, bit lines **3240** electrically connected to the memory channel structures **3220**, and gate contact plugs **3235** electrically connected to the word lines WL (refer to FIG. **14**) of the gate stack structure **3210**. As described above with reference to FIGS. **1** to **9**, each of the semiconductor chips **2200** may include a semiconductor substrate **201**, an active region **205**, an isolation insulating layer **210**, a gate pattern structure GS, a spacer layer **240**, source/drain regions **250**, an upper substrate **101**, a gate electrodes **130**, and channel structures CH.
(105) Each of the semiconductor chips **2200** may include a through-interconnection **3245** electrically connected to the peripheral interconnections **3110** of the first structure **3100** and extending into the second structure **3200**. The through-interconnection **3245** may be outside the gate stack structure **3210** and may penetrate through the gate stack structure **3210**. Each of the semiconductor chips **2200** may further include the I/O pad **2210** (refer to FIG. **15**) electrically

connected to the peripheral interconnections **3110** of the first structure **3100**.
(106) By way of summation and review, a method for increasing data storage capacity of semiconductor devices may include a semiconductor device including memory cells arranged three-dimensionally, instead of memory cells arranged two-dimensionally.
(107) As set forth above, by optimizing the patterning shape of the gate isolation region separating the gate pattern structures, device failure due to rounding of the end portions of the gate pattern structures may be minimized, and short circuits between the gate pattern structures may be prevented by securing the distance between the end portions.
(108) One or more embodiments may provide a semiconductor device having improved electrical characteristics and reliability.
(109) One or more embodiments may provide a data storage system including a semiconductor device having improved electrical characteristics and reliability.
(110) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor device, comprising: a lower structure; and an upper structure on the lower structure and including a memory cell array, wherein: the lower structure includes: a semiconductor substrate, a first active region and a second active region spaced apart from each other in a first direction on the semiconductor substrate, the first active region and second active region being defined by an isolation insulating layer on the semiconductor substrate, and a first gate pattern structure and a second gate pattern structure extending in the first direction to cross the first active region and the second active region, respectively, on the semiconductor substrate, wherein the first gate pattern structure includes: a first gate dielectric layer on the first active region, a first lower gate pattern on the first gate dielectric layer, and a first upper gate pattern on the first lower gate pattern, a side surface of the first gate dielectric layer and a side surface of the first lower gate pattern are substantially coplanar with a side surface of the first active region, and the first upper gate pattern extends in the first direction beyond the side surface of the first lower gate pattern, the first gate pattern structure and the second gate pattern structure have a first end portion and a second end portion spaced apart from each other in a facing manner in the first direction, respectively, and the first end portion and the second end portion are concavely curved in opposite directions away from each other in a plan view.
2. The semiconductor device as claimed in claim 1, wherein: the first gate pattern structure has a first side surface extending in the first direction, the second gate pattern structure has a second side surface extending in the first direction, and a minimum distance in the first direction between the first side surface of the first gate pattern structure and the second side surface of the second gate pattern structure is less than a distance in the first direction between a central portion of the first end portion of the first gate pattern structure and a central portion of the second end portion of the second gate pattern structure in the plan view.
3. The semiconductor device as claimed in claim 1, wherein a distance between the first active region and the second active region in the first direction is about 160 nm or less.
4. The semiconductor device as claimed in claim 1, wherein an upper surface of the isolation

insulating layer between the first active region and the second active region includes a concave region recessed in a direction toward a lower surface of the isolation insulating layer.
5. The semiconductor device as claimed in claim 1, wherein: the second gate pattern structure includes: a second gate dielectric layer on the second active region, a second lower gate pattern on the second gate dielectric layer, and a second upper gate pattern on the second lower gate pattern, a side surface of the second gate dielectric layer and a side surface of the second lower gate pattern are substantially coplanar with a side surface of the second active region, and the second upper gate pattern extends in the first direction beyond the side surface of the second lower gate pattern.
6. The semiconductor device as claimed in claim 1, wherein: the first upper gate pattern includes: a first pattern layer, a second pattern layer on the first pattern layer, and a third pattern layer on the second pattern layer, and the first pattern layer covers a portion of the side surface of the first lower gate pattern and extends along an upper surface of the isolation insulating layer.
7. The semiconductor device as claimed in claim 6, wherein: the first lower gate pattern and the first pattern layer each include a doped silicon layer, the second pattern layer includes a metal nitride layer, and the third pattern layer includes a metal layer.
8. The semiconductor device as claimed in claim 5, wherein: the first gate pattern structure has a first side surface extending in the first direction, the second gate pattern structure has a second side surface extending in the first direction, the first gate pattern structure further includes a first mask pattern layer on the first upper gate pattern, the second gate pattern structure further includes a second mask pattern layer on the second upper gate pattern, and the lower structure further includes: a first spacer layer covering the first side surface and the first end portion of the first gate pattern structure, a second spacer layer covering the second side surface and the second end portion of the second gate pattern structure and spaced apart from the first spacer layer, first source/drain regions in a portion of the first active region on both sides of the first gate pattern structure, second source/drain regions in a portion of the second active region on both sides of the second gate pattern structure, a buffer insulating layer covering the first spacer layer, the second spacer layer, the first source/drain regions, the second source/drain regions, the first mask pattern layer, and the second mask pattern layer, and an etch stop layer on the buffer insulating layer.
9. The semiconductor device as claimed in claim 8, wherein the etch stop layer extends between the first gate pattern structure and the second gate pattern structure and includes a vertex pointing toward an upper surface of the isolation insulating layer between the first active region and the second active region.
10. The semiconductor device as claimed in claim 1, wherein the upper structure includes: an upper substrate on the lower structure, a stack structure including interlayer insulating layers and gate electrodes alternately stacked in a vertical direction, perpendicular to an upper surface of the upper substrate, on the upper substrate, and a channel structure penetrating through the stack structure in the vertical direction and including a channel layer.
11. A semiconductor device, comprising: a lower structure; and an upper structure on the lower structure, wherein: the lower structure includes: a semiconductor substrate, a first active region and a second active region spaced apart from each other in a first direction on the semiconductor substrate, the first active region and second active region being defined by an isolation insulating layer on the semiconductor substrate, and a first gate pattern structure and a second gate pattern structure extending in the first direction to cross the first active region and the second active region, respectively, on the semiconductor substrate, wherein the first gate pattern structure includes: a first gate dielectric layer on the first active region, a first lower gate pattern on the first gate dielectric layer, and a first upper gate pattern on the first lower gate pattern, a side surface of the first gate dielectric layer and a side surface of the first lower gate pattern are substantially coplanar with a side surface of the first active region, and the first upper gate pattern extends in the first direction beyond the side surface of the first lower gate pattern, the first gate pattern structure has a first side surface extending in the first direction and a first end portion facing the second gate pattern

structure, the second gate pattern structure has a second side surface extending in the first direction and a second end portion facing the first end portion of the first gate pattern structure, and a minimum distance in the first direction between the first side surface of the first gate pattern structure and the second side surface of the second gate pattern structure is less than a distance in the first direction between a central portion of the first end portion of the first gate pattern structure and a central portion of the second end portion of the second gate pattern structure in a plan view.
12. The semiconductor device as claimed in claim 11, wherein the first side surface of the first gate pattern structure and a cross-section of the first end portion meet at an acute angle in the plan view.
13. The semiconductor device as claimed in claim 11, wherein the lower structure includes: a first spacer layer covering the first side surface and the first end portion of the first gate pattern structure, and a second spacer layer covering the second side surface and the second end portion of the second gate pattern structure and spaced apart from the first spacer layer.
14. The semiconductor device as claimed in claim 13, wherein: an upper surface of the isolation insulating layer between the first active region and the second active region includes a concave region recessed in a direction toward a lower surface of the isolation insulating layer, and the first spacer layer and the second spacer layer at least partially cover the concave region of the upper surface of the isolation insulating layer.
15. The semiconductor device as claimed in claim 11, wherein at least one of the first active region and the second active region includes an extension portion extending in a second direction, perpendicular to the first direction, and a contact portion on both sides of the extension portion in the second direction and connected to the extension portion, and the contact portion is bent from the extension portion.
16. The semiconductor device as claimed in claim 15, wherein: the contact portion includes a first contact portion and a second contact portion on both sides of the extension portion in the second direction, and the first contact portion and the second contact portion are bent from the extension portion in opposite directions.
17. The semiconductor device as claimed in claim 11, wherein at least one of the first active region and the second active region extends in a direction slanted to the first direction.
18. A data storage system, comprising: a semiconductor storage device including a lower structure including a semiconductor substrate, circuit elements on the semiconductor substrate, and a lower interconnection structure electrically connected to the circuit elements, an upper structure on the lower structure, and an input/output (I/O) pad electrically connected to the circuit elements; and a controller electrically connected to the semiconductor storage device through the I/O pad and controlling the semiconductor storage device, wherein: the circuit elements of the lower structure include: a first active region and a second active region spaced apart from each other in a first direction on the semiconductor substrate, the first active region and second active region being defined by an isolation insulating layer on the semiconductor substrate, and a first gate pattern structure and a second gate pattern structure respectively crossing the first active region and the second active region to extend in the first direction on the semiconductor substrate, wherein the first gate pattern structure includes: a first gate dielectric layer on the first active region, a first lower gate pattern on the first gate dielectric layer, and a first upper gate pattern on the first lower gate pattern, a side surface of the first gate dielectric layer and a side surface of the first lower gate pattern are substantially coplanar with a side surface of the first active region, and the first upper gate pattern extends in the first direction beyond the side surface of the first lower gate pattern, the upper structure includes: an upper substrate on the lower structure, a stack structure including interlayer insulating layers and gate electrodes alternately stacked in a vertical direction, perpendicular to an upper surface of the upper substrate, on the upper substrate, and a channel structure penetrating through the stack structure in the vertical direction and including a channel layer, the first gate pattern structure and the second gate pattern structure include a first end portion and a second end portion spaced apart from each other in a facing manner in the first direction,

respectively, and the first end portion and the second end portion are concavely curved in opposite directions away from each other in a plan view.

19. The data storage system as claimed in claim 18, wherein: the first gate pattern structure has a first side surface extending in the first direction, the second gate pattern structure has a second side surface extending in the first direction, and a minimum distance in the first direction between the first side surface of the first gate pattern structure and the second side surface of the second gate pattern structure is less than a distance in the first direction between a central portion of the first end portion of the first gate pattern structure and a central portion of the second end portion of the second gate pattern structure in the plan view.